

Literature Review — Cross-Dataset Generalization of ML-Based Intrusion Detection

Project: Evaluating and Improving Cross-Dataset Generalization of ML-Based Intrusion Detection Systems **Document:** 02 — Literature Review **Citation style:** IEEE (numbered) **Status:** Prepared for mid-semester review

1. Scope and Method

This review surveys the two bodies of literature that the project connects: (i) studies of **cross-dataset generalization** of machine-learning intrusion-detection systems (IDS), and (ii) **metaheuristic / bio-inspired feature selection** for IDS. It also covers the domain- adaptation literature and the dataset-evaluation-critique literature, because both directly inform how the project should be designed. Papers were retrieved primarily from the arXiv open-access index and the original dataset publications; each is summarised with the same structure: **problem** → **method** → **datasets** → **key results** → **limitation**.

The central finding of the review, expanded in §6, is that the two bodies of work **have not yet been combined**: metaheuristic feature selection for IDS is overwhelmingly evaluated on a *single* dataset, while the cross-dataset community has not used metaheuristics to select transferable features. This intersection is the project’s contribution.

2. Foundational Datasets

[1] I. Sharafaldin, A. H. Lashkari, A. A. Ghorbani (2018) — **CIC-IDS-2017**. Introduces the CICFlowMeter-based CIC-IDS-2017 dataset (78 flow features, realistic enterprise traffic, 14 attack types across five days). It is the de-facto standard for flow-based IDS research, but its identity features (ports) and class imbalance have since been shown to inflate in-dataset accuracy [4], [22].

[2] N. Moustafa, J. Slay (2015) — **UNSW-NB15**. Introduces UNSW-NB15 (Argus + Bro hybrid, 9 attack families), created specifically because older datasets (KDD/NSL-KDD) were outdated. Provides a fixed train/test split (175,341 / 82,332 records) and both a binary label and a 9-class attack_cat.

[3] N. Moustafa (2021) — **TON_IoT**. Describes the TON_IoT heterogeneous IoT/IIoT testbed (edge-fog-cloud, SDN/NFV) and its network dataset, whose features are raw Zeek (Bro) log fields. Its attack classes are near-uniformly balanced, making it a strong *target* domain for transfer studies.

3. Theme 1 — Cross-Dataset Generalization of NIDS

This is the literature the project directly extends.

[4] Cantone, Marrocco, Bria (2024), “*On the Cross-Dataset Generalization of ML for Network Intrusion Detection*,” *IEEE Access* — **most closely related work**

- **Problem:** Do ML-based NIDS generalise when trained on one network and tested on another?
- **Method:** Four classifiers (DT, RF, KNN, MLP) trained and tested across **four** datasets (CIC-IDS-2017, CSE-CIC-IDS2018, LycoS-IDS2017, LycoS-Unicas-IDS2018) using a **restricted common feature representation**; extensive data visualisation to diagnose failures.
- **Key results:** (i) near-perfect same-dataset accuracy; (ii) cross-dataset accuracy drops to roughly random-chance levels for most source→target pairs; (iii) the failures are traced to **dataset-specific artefacts** (identity features such as IPs/ports, and collection methodology) rather than a lack of model capacity.
- **Limitation:** It *diagnoses* the problem but does **not** propose or optimise a solution — it uses a fixed common feature set and standard hyper-parameters. This leaves the “what should we actually do to narrow the gap” question open, which is precisely the project’s contribution.

[5] Hossain et al. (2026), “*Assessing Generalisation Capability of ML Models for Intrusion Detection*” — **direct competitor on two of our three datasets**

- **Problem:** Does strong in-dataset performance imply cross-dataset reliability?
- **Method:** RF, Logistic Regression, and Naïve Bayes evaluated same-dataset and cross-dataset on UNSW-NB15 ↔ TON_IoT (exactly two of our datasets).
- **Key results:** RF reaches 95.08% (UNSW) / 99.79% (TON) in-dataset, but cross-dataset accuracy falls **below 40%** in both directions.
- **Limitation:** Purely observational — it *measures* the gap but performs **no feature selection, no optimisation, and no mitigation**. It also omits CIC-IDS-2017 and any deep/metaheuristic component. Our project subsumes and extends it by adding CIC-IDS-2017, a shared-feature-subspace design, and (crucially) a metaheuristic that *optimises for* cross-dataset robustness.

[6] Hakim, Uddin, Anis (2026), “*Cross-Domain Generalization Failure in Lightweight IDS Models for IIoT*”

- **Problem:** Do lightweight models trained on one IIoT dataset transfer to structurally different IIoT datasets?
- **Method:** Four lightweight models trained on one IIoT source and tested (no retraining) on two targets using a feature representation restricted to attributes shared by all three; explainability (SHAP-style) analysis; adversarial-robustness and limited-adaptation probes.
- **Key results:** (i) both top models rely overwhelmingly on **coarse port-category features**, which occur 96–435× more often in the source attack traffic than in targets — i.e. a **shortcut**; (ii) the evaluation protocol (class-imbalance handling) can **reverse** which target appears harder; (iii) adversarial robustness is *unrelated* to cross-network generalisation.
- **Limitation:** Like [4]/[5], it diagnoses; it does not select features *for* transfer. **Directly informs our design:** we must strip port/identity features and evaluate under realistic class distributions.

[7] Butt, Hotho, Schlör (2026), “*Evaluating Tabular Representation Learning for NIDS*”

- **Problem:** Can representation learning replace hand-crafted NetFlow features and improve transfer?

- **Method:** Systematic evaluation of SOTA tabular representation-learning methods (incl. TabICL), autoencoders, and end-to-end transformers on several NetFlow datasets, with cross-dataset transfer experiments.
- **Key results:** strong **dataset–model dependency** (no method dominates everywhere); cross-dataset transfer works *sometimes* but varies greatly by source–target pair.
- **Limitation:** Representation-level, not feature-selection-level; no optimisation for transfer. Supports our claim that the *choice of what to feed the model* matters as much as the model.

[31] Pontes et al. (2021), “A New Method for Flow-based NIDS Using the Inverse Potts Model”

- **Problem:** Classical ML classifiers overfit a single distribution and lack domain adaptation.
- **Method:** Energy-based Flow Classifier (EFC), an anomaly detector trained only on benign flows via inverse statistical modelling.
- **Key results:** EFC is **more adaptable across distributions** than supervised baselines on CIDDs-001, CICIDS17, and CICDDoS19.
- **Limitation:** Binary anomaly detection only; not a supervised multi-class transfer solution. Relevant as evidence that *distribution-agnostic* representations transfer better.

[30] Xu, Liu (2025), “Robust Anomaly Detection in Network Traffic: Evaluating ML Models on CICIDS2017”

- **Problem:** How do supervised vs unsupervised models generalise to *unseen attack types* within a dataset?
- **Method:** MLP, 1D-CNN, OCSVM, LOF on CICIDS2017 under “known attacks” vs “novel attacks” scenarios.
- **Key results:** supervised MLP/CNN are near-perfect on familiar attacks but collapse on novel ones; boundary-based OCSVM trades off precision/recall best.
- **Limitation:** Within-dataset novelty, not cross-dataset. Reinforces that supervised models are fragile to distribution shift — the same effect we study across datasets.

4. Theme 2 — Domain Adaptation and Transfer Learning for NIDS

This literature shows the “sophisticated transfer” route, against which a *feature-selection* route can be compared.

[8] Layeghy, Baktashmotlagh, Portmann (2022/23), “DI-NIDS: Domain Invariant NIDS”

- **Method:** Adversarial domain adaptation (DANN-style) to learn domain-invariant features from labelled sources, then a One-Class SVM on the invariant space.
- **Datasets/results:** NFv2-CIC-2018 and NFv2-UNSW-NB15; superior cross-domain performance vs prior DA approaches.
- **Limitation:** Requires aligned (NetFlow v2) feature spaces; unsupervised detection only.

[14] Wu et al. (2022), “*JSTN: Joint Semantic Transfer Network for IoT Intrusion Detection*”

- **Method:** Multi-source heterogeneous domain adaptation transferring three “semantics” (scenario, weighted-implicit, hierarchical-explicit alignment) from a network-intrusion source to a data-scarce IoT target.
- **Results:** ~10.3% average accuracy boost vs SOTA baselines (IEEE IoT-J).
- **Limitation:** Targets *data-scarcity* (few labelled IoT samples), a different motivation than our *generalisation-under-schema-mismatch* problem.

[15] Wu et al. (2023), “*GGA: Heterogeneous DA via Geometric Graph Alignment*”

- **Method:** Aligns intrusion *category graphs* (vertices = categories, edges = relations) between domains to transfer knowledge despite heterogeneous feature spaces.
- **Limitation:** Requires overlapping *category* semantics; heavy machinery; evaluated for IoT-target transfer.

[16] Wu et al. (2023), “*ABRSI: Adaptive Bi-Recommendation and Self-Improving Network*”

- **Method:** Unsupervised heterogeneous DA with recommender-style bi-matching, hybrid pseudo-labelling, and error-knowledge learning.
- **Limitation:** Same family of data-scarcity transfer; no feature-selection interpretation.

[17] Wu et al. (2023), “*OSDN: Open-Set Dandelion Network*”

- **Method:** Open-set heterogeneous DA; “dandelion” feature-space shaping for inter-class separability; detects target classes unseen in the source.
- **Results:** Outperforms three SOTA baselines by 16.9% (ACM ToIT).
- **Limitation:** Open-set and label-scarce focus; not a benchmark of *classical* models under schema mismatch.

[18] Varghese, Taghiyarrenani (2025), “*Intrusion Detection in Heterogeneous Networks with Domain-Adaptive Multi-Modal Learning*”

- **Method:** Deep model integrating multi-modal learning with DA, processing multiple datasets cyclically to adapt to varying feature spaces.
- **Limitation:** Requires joint access to multiple datasets; black-box.

[19] Wang (2026), “*Clustering-Enhanced Domain Adaptation for Cross-Domain IDS in ICS*”

- **Method:** Spectral-transform feature alignment + K-Medoids/PCA clustering enhancement for industrial-control traffic.
- **Results:** up to +49% accuracy vs five baselines; clustering adds up to +26%.
- **Limitation:** ICS-specific; still requires target-domain exposure (transductive).

[20] Nguyen, Watabe (2023), “*Flow-Sequence NIDS via BERT*”

- **Method:** Encode *sequences of flows* with a BERT transformer to improve domain adaptation.
- **Limitation:** Needs ordered flow sequences (not plain tabular rows); early-stage results.

[33] Hossain et al. (2026), “*Cross-Dataset Zero-Day IDS via Siamese Network + RL*”

- **Method:** Siamese network for anomaly correlation + PPO reinforcement-learning adaptive defence for zero-day detection.
- **Results:** 99.07% unknown-attack accuracy, 93.94% zero-day ratio on IoT benchmarks.
- **Limitation:** Unsupervised/anomaly framing; not a reproducible cross-dataset *benchmark* with classical models; no feature-selection lens.

Summary of Theme 2: domain adaptation works, but (a) it usually assumes overlapping label semantics and aligned or learnable-aligned feature spaces, (b) it requires target-domain data, and (c) it is model-centric. **None of these works use metaheuristic feature selection**, and none frame the problem as “choose a small transferable feature subset evaluated under a strict train-on-source/test-on-target protocol.”

5. Theme 3 — Metaheuristic / Bio-inspired Feature Selection for IDS

This is the “optimisation” literature the project uses as its engine.

[9] Fatima, Ali (2022), “*Effective Metaheuristic-Based Classifiers for Multiclass IDS*”

- **Method:** Wrapper **Genetic Algorithm (GA)** feature selection + stacking/bagging ensembles on UNSW-NB15 and CICDDoS2019.
- **Results:** GA + stacking improves multi-class accuracy significantly; better DR and lower FAR than baselines.
- **Limitation:** Fitness = **same-dataset** accuracy. The selected subset is optimised for the *training* distribution, so nothing guarantees transfer. **This is the exact pattern the project reverses** (we optimise for cross-dataset robustness instead).

[10] Suman, Tripathy, Saha (2019), “*Building an Effective IDS using Unsupervised Feature Selection in a Multi-objective Optimization Framework*”

- **Method:** **NSGA-II** optimising three information-theoretic objectives (mutual information, std-dev, information gain) for *unsupervised* feature selection; Pareto-optimal subsets feed SVM/DT/KNN.
- **Results:** 99.78% accuracy on KDD-99, 99.83% on NSL-KDD (20% test), 99.65% on Kyoto.
- **Limitation:** Single-dataset evaluation on legacy datasets; the multi-objective idea (accuracy vs #features) is useful for us, but our objectives will be *cross-dataset*.

[11] Li (2026), “*MPDGGA: Multi-population Diversity-guided GA for Feature Selection in NIDS*”

- **Method:** Chained multi-population GA with an information-gain-ratio diversity operator.
- **Results:** Best accuracy on 10/11 datasets (NSL-KDD, UNSW-NB15, 9 UCI) while selecting $\geq 2.26\%$ of features.
- **Limitation:** Still single-dataset accuracy; no cross-dataset protocol.

[12] Mojtabedi et al. (2022), “*FS-based IDS using Genetic Whale Optimization + Sample-based Classification*”

- **Method:** Hybrid WOA + GA feature selection with KNN on KDDCUP1999.
- **Limitation:** Legacy dataset, single-domain, accuracy-only fitness.

[13] Yang et al. (2026), “*Multi-Objective AutoML-based Efficient IDS for EV Charging Networks*”

- **Method:** NSGA-III jointly optimising LightGBM feature threshold + hyper-parameters under three objectives (weighted F1, inference latency, model size).
- **Datasets/results:** CICEVSE2024 and CICIDS2017; competitive F1 with lower latency/size.
- **Limitation:** Deployment-efficiency objectives, not generalisation; single-dataset.

[32] Barati (2025), “*Quantum GA-enhanced Self-Supervised IDS for WSN/IoT*”

- **Method:** Quantum-inspired GA for feature selection + self-supervised representation learning.
- **Limitation:** Niche (quantum-inspired); single-dataset IoT evaluation.

Summary of Theme 3: the metaheuristic FS literature is mature and effective, but its fitness functions are uniformly **same-dataset accuracy/F1**. To our knowledge, **no published work optimises a feature subset for cross-dataset generalisation**. This is the gap.

6. Theme 4 — Dataset & Evaluation Critique (why careful protocol matters)

- [25] Ring et al. (2019), “**A Survey of Network-Based Intrusion Detection Data Sets**”: canonical taxonomy of NIDS datasets and their properties; documents that datasets differ in collection environment, feature semantics, and label quality — the root of the generalisation problem.
 - [22] Vitorino et al. (2024), “**An Adversarial Robustness Benchmark for Enterprise NIDS**”: shows a *corrected* CICIDS2017 (“NewCICIDS”) changes model rankings; underlines that the dataset’s known flaws materially affect results.
 - [23] Youssef (2026), “**Stream Assembly Is an Uncontrolled Treatment in Streaming IDS Benchmarks**”: demonstrates that *how* evaluation streams are assembled (interleaving, pooling, replay) changes what is measured and can reverse rankings — a warning we take seriously for our train/test protocol.
 - [21] Corsini, Yang, Apruzzese (2021), “**On the Evaluation of Sequential ML for NIDS**”: establishes fair-comparison methodology for flow-based models; motivates our decision to use matched, reproducible splits and macro metrics.
 - [24] El Mahdaouy et al. (2026), “**Deep Learning for Contextualized NetFlow-based NIDS**”: survey that lists **weak cross-dataset generalization** among the key failure modes that inflate reported results — direct justification for our benchmark design.
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7. Synthesis and Research Gap

The following table summarises the gap. “Cross-dataset evaluation” means a strict train-on-source / test-on-target protocol; “metaheuristic FS” means GA/PSO/NSGA-style feature selection; “optimises for transfer” means the fitness function targets cross-dataset performance rather than same-dataset accuracy.

Work	Datasets ($n \geq 3$, hetero- geneous)	Cross- dataset eval	Metaheuris- tic FS	Optimises for transfer	Mitigates gap
Cantone et al. [4]	✓ (4)	✓	✗	✗	✗ (diagnosis only)
Hossain et al. [5]	~ (2)	✓	✗	✗	✗
Hakim et al. [6]	✓ (3 IIoT)	✓	✗	✗	✗
Butt et al. [7]	✓	✓	✗	✗	partial (representation)
DI-NIDS [8] / JSTN [14] / GGA [15] / OSDN [17]	✓	✓ (DA)	✗	✗ (aligns features, not selects)	partial (needs target data)
GA/PSO/NSGA FS [9]–[13], [32]	✗ (single)	✗	✓	✗ (same-dataset fitness)	✗
This project	✓ (3, heterogeneous)	✓	✓ (GA, PSO, +)	✓ (cross-dataset fitness)	✓ (goal)

7.1 Explicit gap statements

- Gap 1 (primary):** No published work applies **metaheuristic/bio-inspired feature selection whose fitness is cross-dataset generalisation** to network intrusion detection. All existing metaheuristic FS optimises same-dataset accuracy [9]–[13]; all existing cross-dataset work is either diagnostic [4]–[6] or model-centric domain adaptation [8], [14]–[20].
- Gap 2:** There is no **systematic benchmark** that jointly evaluates the *same* ML/DL models across **CIC-IDS-2017, UNSW-NB15, and TON_IoT** under a shared feature subspace and reports which *features* and *attack classes* transfer versus which are dataset artefacts.
- Gap 3:** No work consolidates these findings into **practical, feature-level guidelines** for building IDS that generalise to unseen networks.

These three gaps map one-to-one onto the project’s five objectives (see 03_objectives_and_gaps.md).

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